

Computing Methods for Physics 1 – 31 January 2024

Your exam material (code files, plots, datafiles, etc.) must be submitted via google classroom by 14:15 as a single zip file.

C++ evaluation will be based on: correct syntax, proper return types, proper arguments of functions, data members and class interfaces, setters/getters, unnecessary void functions, correct mathematical expressions, comments throughout the code, separation of class implementations and interfaces.

Python evaluation will be based on: correct syntax, avoiding C-style loops, using Python features in general, comments throughout the notebook/scripts, labels, legends and plot styling and clarity in general.

Part 1 – C++

Implement a class to represent and handle polynomials of a single variable (x) in the domain of real numbers. The class must provide the following.

- A regular constructor and a copy constructor.
- A `print` method.
- A method to evaluate the polynomial at a specific value of the single variable.
- A method to determine the i -th derivative of the polynomial with respect to the single variable.
- A method to calculate the integral of the polynomial over a finite interval $[a, b]$.
- Overloaded `+`, `-`, and `*` operators to add, subtract, and multiply two polynomials or a polynomial and a (real) number.
- An overloaded version the `==` operator to determine if two polynomials are the same.

Part 2 – Python

Use Python to simulate the decay of a particle into lighter particles. The decaying particle has rest mass M and lifetime τ and it travels along the x -axis (of the laboratory frame) with velocity v_0 . The decay occurs at a time $\bar{t}$.

- Write a class to represent the decay. The rest mass and lifetime of the decaying particle must be two attributes of the class. A third mandatory attribute is a collection to be used to specify the decay products (when the user fills this up, it will contain other instances of the `decay` class). You must supply the class with a print method.
- Write a method that relies on `numpy.random.exponential` to draw the decay time $\bar{t}$ from an exponential distribution based on τ .
- Write a method to assign the decay products their momentum in the case of two particles being produced. In this scenario, in the center of mass of the decaying particle, the decay products have momentum

$$p^* = \frac{\sqrt{M^4 + m_1^4 + m_2^4 - 2M^2m_1^2 - 2M^2m_2^2 - 2m_1^2m_2^2}}{2M}c,$$

where m_1 and m_2 are their rest masses, and their direction is random. Do not forget to boost the 4-momenta $p^\mu = (E/c, p_x, p_y, p_z) = m(c, v_x, v_y, v_z)/\sqrt{1 - v^2/c^2}$ to the laboratory frame.

- Set up the simulation of a $B^0 \rightarrow K^+ + \pi^-$ decay event by exploiting your class and asking the user for v_0 . [$m_{B^0} = 5280 \text{ MeV}/c^2$, $\tau_{B^0} = 1.5 \text{ ps}$, $m_{K^+} = 494 \text{ MeV}/c^2$, $\tau_{K^+} = 12 \text{ ns}$, $m_{\pi^-} = 140 \text{ MeV}/c^2$, and $\tau_{\pi^-} = 26 \text{ ns}$.]
- Generate a 3D plot of the simulated decay, from $t = 0$ to $t = 2\bar{t}$.